

Shengyi Guan

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Research Interests

Topological quantum field theory; generalized and categorical symmetries; geometric representation theory; derived and enumerative geometry; Calabi–Yau geometry and mirror symmetry.

Education

Peking University, School of Physics

Aug 2024–Jun 2028

B.S. in Physics, Outstanding Talent Training Program in Physics

Expected

GPA: 3.887; ranked in the top 5% of the cohort after the first two academic years.

Selected graduate coursework: Group Theory I (A+), Quantum Field Theory (A-), and General Relativity (96/100).

Research Experience

Undergraduate Researcher, Peking University

May 2025–Present

Advisor: Prof. Yinan Wang

Beijing, China

- Investigating the large-level limit of doubled Chern–Simons theory toward BF theory, with emphasis on categorical data on the circle and the regular sector.
- Studying a candidate limit $\mathcal{Z}(\mathrm{Rep}^k(LG)) \rightarrow \mathrm{Rep}(C^*(G//G))$ using geometric quantization, representation theory, and operator-algebraic methods.
- Research conducted within the Peking University Undergraduate Research Training Program.

Manuscript in Preparation

Shengyi Guan (co-first author), Qiang Jia, Zhongxuan Li, Jiahua Tian, and Yinan Wang. “From Doubled Chern–Simons to BF Theory: A Regular-Sector Limit of Categorical Data.” Manuscript in preparation, Aug 2026.

Selected Academic Preparation

Physics preparation includes quantum field theory, gauge theory, string theory, and two-dimensional conformal field theory. Mathematical preparation includes topology, differential geometry, category theory, homological algebra, sheaf theory, derived categories, and stable ∞ -categories.

Honors and Awards

- Silver Medal (4th place nationally)**, Mathematical Physics, 16th S.-T. Yau College Student Mathematics Contest, 2025.
- National Second Prize**, Final Round, 40th Chinese Physics Olympiad (CPhO), 2023.
- National Second Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), 2025.
- First Prize**, Mathematics-Major Division, 36th Beijing Undergraduate Mathematics Competition, 2025.
- Peking University Qinyuan–Shizi Yunhui Scholarship, 2024–2025.
- Peking University Merit Student Honor (Sanhao Student), 2024–2025.
- Second Prize**, Non-Mathematics-Major Division, 35th Beijing Undergraduate Mathematics Competition, 2024.
- Second Prize**, Physics Division, 41st Undergraduate Physics Competition for Selected Regions of China, 2025.